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Research Article

Unveiling Solubilization Mechanisms of Natural Deep Eutectic Solvents in Triazole Fungicides: COSMO-RS Calculations and Screening for Eco-Friendly, High-Efficiency Pesticide Systems

Juan Wang, Fengmao Liu,* Qingrong Peng,* Xianzhao Zhang, Zhiqiang Gao, Peirong Zhang, Wenzhuo Wang, and Jing Sun

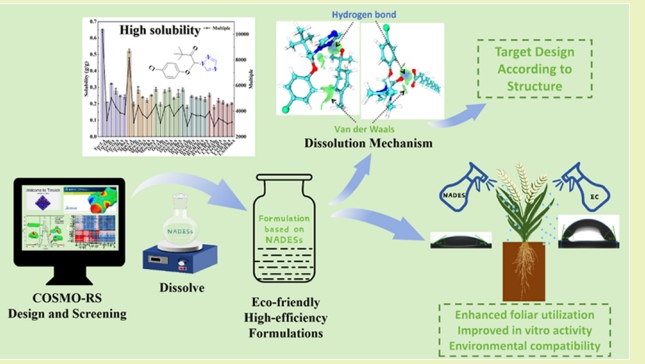
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ABSTRACT: Enhancing the dissolution of lipophilic agrochemicals is paramount for addressing agricultural environmental pollution and improving utilization efficiency. This study introduced a novel methodology to enhance pesticide solubility by integrating theoretical calculations, targeted design, and high-throughput screening for the development of sustainable solvents. Natural deep eutectic solvents (NADESs) optimized for pesticide formulation were systematically designed and selectively filtered from 1768 species, exhibiting exceptional solubility for triazole fungicides identified through the conductor-like screening model for real solvents (COSMO-RS). Experimental results demonstrated a significant increase in solubility relative to water, ranging from 2.8 to 3.6 × 10^5 times. Molecular dynamics simulations, quantitative calculations, and property measurements elucidated van der Waals and hydrogen bonding interactions as the dominant roles of the solubilization mechanisms. Furthermore, the novel eco-friendly pesticide system, integrating triadimefon with three NADESs, exhibited a 0.87 times increase in inhibitory activity against *Fusarium graminearum*, accompanied by a 15.6% reduction in equilibrium surface tension and a 58.1% decrease in contact angle on wheat leaves compared to commercial emulsifiable concentrate (EC). These findings underscore the potential of NADESs-based formulations to enhance pesticide utilization efficiency with minimal impact on wheat growth indicators, higher cell survival, and increased biocompatibility, thus providing innovative strategies for sustainable agricultural design.



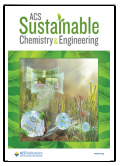
KEYWORDS: *lipophilic agrochemicals, solubilization mechanisms, COSMO-RS modeling, molecular dynamics simulation, NADESs, eco-friendly pesticide formulations*

INTRODUCTION

Pesticides play a pivotal role in mitigating crop pests and diseases, enhancing crop yield, and supporting modern agricultural development.[1−3] Despite their significant contributions, the high hydrophobicity of pesticide active ingredients (AIs) necessitates the extensive use of organic solvents in both manufacturing and application processes, resulting in substantial volatile organic compounds (VOCs) emissions.[4,5] Studies indicated that agricultural sources contribute to approximately 14% of anthropogenic VOCs emissions.[1] The inherent challenges of poor water solubility and low foliar retention rate of active ingredients compel increased usage, leading to elevated levels of aromatic hydrocarbon discharge into the environment and subsequent environmental pollution and ecological damage.[6−8] Hence, there is a critical need to design novel, environmentally friendly, naturally derived, and tunable green solvents. Moreover, the development of sustainable pesticide formulation systems is imperative to curb organic solvent emissions, optimize pesticide utilization, and enhance overall environmental safety and human health.

The identification of deep eutectic solvents (DESs) represents a significant advancement in green chemistry. DESs, a groundbreaking solvent system initially proposed by Abbott et al.[9] in 2003, are characterized as low-melting-point mixtures with specific stoichiometric ratios of hydrogen bonding donors (HBDs) and hydrogen bonding acceptors (HBAs), connected through hydrogen bonding. These mixtures exhibit considerably lower melting points than either individual component, demonstrating high thermal stability,

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low volatility, low vapor pressure, design flexibility, cost-effectiveness, biodegradability, and biocompatibility.[10] In 2011, Choi et al.[11] extended this concept to include solvents derived from natural primary metabolites of organisms, such as amino acids, organic acids, sugars, choline derivatives, or urea, terming them natural deep eutectic solvents (NADESs). NADESs, recognized as "designer solvents," are environmentally friendly, pose lower toxicity risks, and find widespread applications across various fields.[12−16] Notably, NADESs exhibit significant potential in enhancing drug solubility,[17−19] permeability,[20,21] and stability[19,22,23] due to their exceptional solubilization capabilities across a broad spectrum of compounds.

Despite the well-established applications of NADESs in pharmaceuticals, their utilization in pesticide formulation remains at an early stage of exploration. Dunan et al.[24] investigated formulations containing malic acid/sorbitol/water 1:1:3 and malic acid/serine/water 1:1:3, showcasing high solubility to the phytocide ethylene oxide. These formulations, when used as essential oils, not only minimized damage to the target plant but also exhibited increased tobacco louse mortality and more gradual release kinetics, suggesting the potential for effective and sustained pest control. In a separate study, Wang et al.[25] designed a system involving DESs and mesoporous silica nanoparticles, demonstrating that DESs can influence the arrangement and spatial structure of lignin and cellulose. This modulation enhances the leaf barrier function, improves permeability, and facilitates the distribution and transport of pesticides within the plant. Additionally, prior investigations have confirmed the robust solubilizing capacity of quaternary DESs for highly hydrophobic pesticide AIs.[26] These findings introduce innovative possibilities for NADESs to replace hazardous solvents in traditional pesticide formulations, paving the way for the development of green and efficient pesticide-like systems.

An inherent feature of NADESs is their designability, a characteristic that broadens their application spectrum while simultaneously introducing increased complexity and selection costs. The conductor-like screening model for real solvents (COSMO-RS) model, introduced by Klamt et al.,[27] stands out as a predictive model amalgamating quantum chemistry and statistical thermodynamics. This model quantifies thermodynamic properties and associated parameters without relying on experimental data, making it a valuable tool for solvent design and screening across various domains.[28−32] The model's reliability in predicting outcomes not only mitigates the need for extensive trial-and-error approaches but also enhances research efficiency and accuracy.

Triazole fungicides (TFs) have gained widespread application in agriculture and industry, constituting 30% of the global fungicide market, attributed to their exceptional characteristics encompassing broad-spectrum efficacy, high efficiency, prolonged persistence, and commendable systemic properties.[33] However, the intrinsic ultralow solubility in water necessitates the formulation of TFs into conventional dosage forms such as emulsifiable concentrate (EC). In these formulations, the inclusion of cosolvents not only introduces their own contaminants but also has the potential to elevate environmental toxicity,[34,35] reproductive metabolism toxicity,[36,37] and induce endocrine-disrupting effects[38,39] associated with TFs. These concerns pose significant threats to human health and environmental well-being.

In this investigation, six TFs (Table S1) served as the target compounds. Specifically, for triadimefon (TDF), the systematic design of NADESs involved a comprehensive selection of natural constituents, including organic alcohols, terpenes, organic bases, organic amines, amino acids, and other pertinent components, utilizing the COSMO-RS methodology. The identification of suitable NADESs was facilitated by evaluating their solubility capacities, with the reliability of the simulation calculations validated through experimental solubility measurements. Furthermore, the solubilization mechanism of TDF by NADESs was thoroughly elucidated by integrating COSMO-RS with molecular dynamics (MD) simulations. Additionally, the applicability of the designed screening approach was validated by extending the solubility measurements to five other TFs: myclobutanil, hexaconazole, tebuconazole, difenoconazole, and propiconazole. Subsequently, three environmentally friendly formulations of TDF based on NADESs were developed and systematically compared with commercial EC concerning in vitro activity, foliar adhesion effects, biocompatibility, and cytotoxicity.

MATERIALS AND METHODS

Materials. The components of NADESs involved in this study shown in Table S2 (purity, 98%) were procured from Sigma-Aldrich Trading Co., Ltd. (Shanghai, China). Six TF technicals (purity, 97%) were sourced from Changshu Hengyao New Material Co., Ltd. (Jiangshu, China). Methanol and acetonitrile (chromatography grade) were acquired from Meridian Technology Co., Ltd. (Beijing, China).

Dissolution and Mechanism Exploration. *Design and Screening of Solvents.* The screening of NADESs was conducted in conjunction with the COSMO-RS design. Initially, TURBOMOLE software (version 21.0.0) was utilized, with def-TZVP chosen as the computational basis set and the computational temperature set at 25 °C. Density functional theory (DFT) calculations were employed to optimize the TFs AIs to their lowest energy states, generating corresponding energy files (molecular structures of TFs and NADESs components were retrieved from the PubChem database). Subsequently, single-point calculations were performed by using the COSMO continuous solvatization method. The surface shielding charge (σ) density distribution was calculated, and the relevant information was stored in a cosmo file, applicable in BIOVIA COSMOtherm software (version 22.0.0). The COSMOtherm software contains a COSMObase with components of NADESs. Utilizing the "multisolvent" panel in COSMOtherm, the solubility of target pesticides in the designed NADESs was calculated by determining the molar ratios of the components. Additionally, COSMOtherm computed the fusion properties of individual components of the screened NADESs based on the solubility parameter. These properties were utilized in predictive modeling of solid−liquid equilibrium (SLE) plots by COSMOtherm, determining the eutectic melting temperature as a crucial parameter for assessing the feasibility of NADESs preparation under room-temperature conditions.

Preparation of NADESs. Preparation of NADESs was conducted using a heating and stirring method.[40,41] Following the COSMOtherm-designed and screened NADESs combinations, components were meticulously weighed in a 1:1 molar ratio and mixed in a 250 mL round-bottomed flask. The mixture was then subjected to heating and stirring under a water bath at 70−80 °C, facilitated by a magnetic stirrer (DF-101S, China), until a clarified and transparent liquid was achieved. Subsequently, the solution was cooled to room temperature, allowing it to stand for 24 h while maintaining its clarified and transparent state, indicating the successful preparation of NADESs.

Determination of Experimental Solubility. Actual solubility assessments were conducted by employing the saturated solution method. Specifically, 1 g of NADESs was precisely weighed into a 2 mL plastic centrifuge tube. The appropriate quantity of the TFs

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technicals was added incrementally to the NADESs, followed by 5 min of vortexing to ensure homogeneous dispersion and solubilization of the pesticides. The resulting mixture underwent ultrasonic (SY-500MD, China) treatment at room temperature for 30 min, promoting further dissolution until saturation was achieved. Subsequently, the solution underwent centrifugation (HC-2517, China) at 10,000 rpm for 1 min, and the supernatant was filtered through a 0.22 μm organic membrane. It was then subjected to high-performance liquid chromatography (Agilent 1260) (HPLC) for the determination of actual solubility, and analytical methods are described in Table S1.

Characterization of NADESs. *Viscosity.* The viscosity of the three most excellent dissolving ability NADESs (terpineol/decanoic acid 1:1, menthol/decanoic acid 1:1, and decyl alcohol/decanoic acid 1:1) was measured using a Brookfield DV1 viscometer (Bleafy). Thirty milliliters of each NADESs was placed in 50 mL plastic centrifuge tubes and subjected to a thermostatic water bath for 30 min. Viscosity measurements were then conducted at temperatures in the range of 20−60 °C. Each measurement was repeated three times.

Density. The density of three NADESs was measured using a thermostatic liquid densitometer (DH-300L-T, China) over a temperature range of 20−60 °C. Each treatment was repeated three times.

Refractive Index. The refractive index (n_D) of the three NADESs was measured by using an Abbe refractometer (ABBE-3L) with a yellow beam at a wavelength of 589 nm. Measurements were taken at 25 ± 0.5 °C, and each treatment was repeated five times. The molar refraction was obtained from eq 1.

$$R_D = \left(\frac{M}{\rho}\right)\left(\frac{n_D + 1}{n_D^2 + 1}\right) \quad (1)$$

where R_D, M, ρ, and n_D are molar refraction (cm^3/mol), molar mass (g/mol), density (g/cm^3), and refractive index, respectively.

Thermal Stability. The thermal properties of the NADESs were assessed using a differential scanning calorimeter (DSC) (NETZSCH, Germany). Approximately 10 mg of each sample was placed in an aluminum crucible and maintained at a constant temperature for 10 min under a nitrogen atmosphere. The temperature was then increased from −90 to 200 °C at a constant rate of 10 °C/min. Each treatment was repeated three times.

Polarity. The Kamlet−Taft polarity parameters of the NADESs were determined using a UV−vis spectrophotometer (TU-1901, China).[42] A solution comprising 0.05 mmol/L 4-nitroanisole, 0.05 mmol/L 4-nitroaniline, and 1 mmol/L Nile Red was initially prepared in methanol. Subsequently, 50 μL of the dye solution was added to a 2 mL centrifuge tube and dried under vacuum at 40 °C for 6 h to remove methanol. Following this, 1.0 g of NADESs was added to the centrifuge tube, vortexed to dissolve the dye, and ensured homogeneity with the staining dye. The adsorption spectra were then determined by using a UV−vis spectrophotometer. The Kamlet−Taft solute discoloration parameter was obtained from eqs 2 to 5.

$$\pi^* = 14.57 - 4270/\lambda_{max,OMe} \quad (2)$$

$$\beta = 11.134 - 3580/\lambda_{max,NH_2} - 1.125\pi^* \quad (3)$$

$$\alpha = (19.9657 - 1.0241 \times \pi^* - \nu_{NR})/1.6078 \quad (4)$$

$$\nu_{NR} = 1/(\lambda_{max,NR} \times 10^{-4}) \quad (5)$$

where π^*, α, and β denote the polarizability, hydrogen-giving capacity, and hydrogen-receiving capacity of the solvent, respectively. And the $\lambda_{max,OMe}$, λ_{max,NH_2}, and $\lambda_{max,NR}$ represent the maximum absorption wavelengths of 4-nitroanisole, 4-nitroaniline, and Nile red, and ν_{NR} represents the maximum absorption wavenumber of Nile red.

Dissolution Mechanism Studies. The dissolution mechanism was elucidated through MD simulations using GROMACS 2021.6 software suite.[43,44] The GROMOS96 54a7 United-Atom force field was employed to describe all molecules,[45] and the parameters for these molecules were sourced from the Automated Topology Builder Web site.[46,47] TDF was selected as the model compound for the study, and MD simulations were conducted for three different ratios of the terpineol and decanoic acid solvent components. See Supporting Files for specific operations and parametrization of solvent simulation. The free energies of solvation of TDF in these three solvents were subsequently calculated. Quantum chemical calculations were carried out using the ORCA-5.0.4 program.[48] Geometry optimization and frequency calculations, as well as single-point energy calculations for each optimized geometry, were performed at higher energy levels employing wB97M-V/aug-cc-pVTZ to attain more accurate interaction energies and facilitate wave function analysis.[49] All electronic structure analyses were conducted using Multiwfn 3.8 (dev) program.[50] The energy and wave function analyses for visualizing the structures were performed using the VMD program.[51]

Application of TDF-NADESs Formulations. *Preparation and Stability Assessment of TDF-NADESs.* Referring to commercial EC formulations, TFs based on NADESs formulations with varying concentrations were separately prepared. The storage stability at elevated temperature of the AIs in six TFs was evaluated following the CIPAC MT46.3. These formulations were sealed in glass bottles and subjected to thermal storage in a constant temperature chamber at 54 ± 2 °C for 14 days. Subsequently, the samples were retrieved, placed in a desiccator, and cooled to room temperature, and the mass fraction of the AIs was determined.

In Vitro Activity Assessment. The in vitro inhibitory activity of TDF-NADES systems against *Fusarium graminearum* was evaluated using the mycelial growth rate method. For instance, the formulation of 15% TDF based on three NADESs (terpineol/decanoic acid 1:1, menthol/decanoic acid 1:1, and decyl alcohol/decanoic acid 1:1) was prepared, with the 25% TDF commercial EC serving as a reference. *F. graminearum* cake (4 mm diameter) was inoculated into the center of potato dextrose agar (PDA) medium containing varying concentrations of TDF formulation, and sterile water was used as control treatment. The PDA medium was incubated at 28 ± 0.5 °C, and the mean diameter of mycelial growth was measured using the criss-cross method at 24, 48, and 72 h of incubation, respectively. Each treatment was replicated three times. The EC_{50} was calculated by using the probit-log method. The inhibitory activity was expressed as mycelial growth inhibition (IR), which was obtained by eq 6.

$$IR\,(\%) = (D_0 - D_t)/(D_0 - 4) \times 100 \quad (6)$$

where IR is mycelial growth inhibition. D_0 represents the diameter of control colonies (mm), D_t represents the diameter of treatment colonies (mm), and 4 represents the diameter of *F. graminearum* cake (mm).

Wheat Leaf Adhesion. Fresh wheat leaves were chosen as the substrates for testing. The dynamic contact angle (DCA) of TDF-NADESs and the commercial EC on wheat leaves were measured using a contact angle meter (OCA50, Germany). The static contact angle at 120 s was considered the equilibrium contact angle of the droplets from the sample solution. Each experiment was conducted in quintuplicate. The equilibrium surface tension (EST) of TDF-NADESs and the EC solutions at various dilutions was determined using the Wilhelmy plate method with a tensiometer (DCAT 21, Germany). Each sample was analyzed in quintuplicate. The surface structure of wheat blades was characterized by using a field emission scanning electron microscope (SEM) (SU8100, Hitachi). To comprehensively assess adhesion and spreading abilities, the adhesion function of different solutions on wheat leaves was calculated based on the wetting equation, incorporating average values of EST at a 1000-fold dilution and the contact angle of droplets on wheat leaves at 120 s. The adhesion (W_a) was obtained from eq 7.

$$W_a = \gamma(1 + \cos\theta) \quad (7)$$

where W_a is the work of adhesion (J/m^2), γ is the EST of the pharmaceutical system (mN/m), and θ is the contact angle (deg).

Assessment of Wheat Safety. Wheat seeds were soaked in distilled water for 4 h and then placed in Petri dishes (d = 15 cm)

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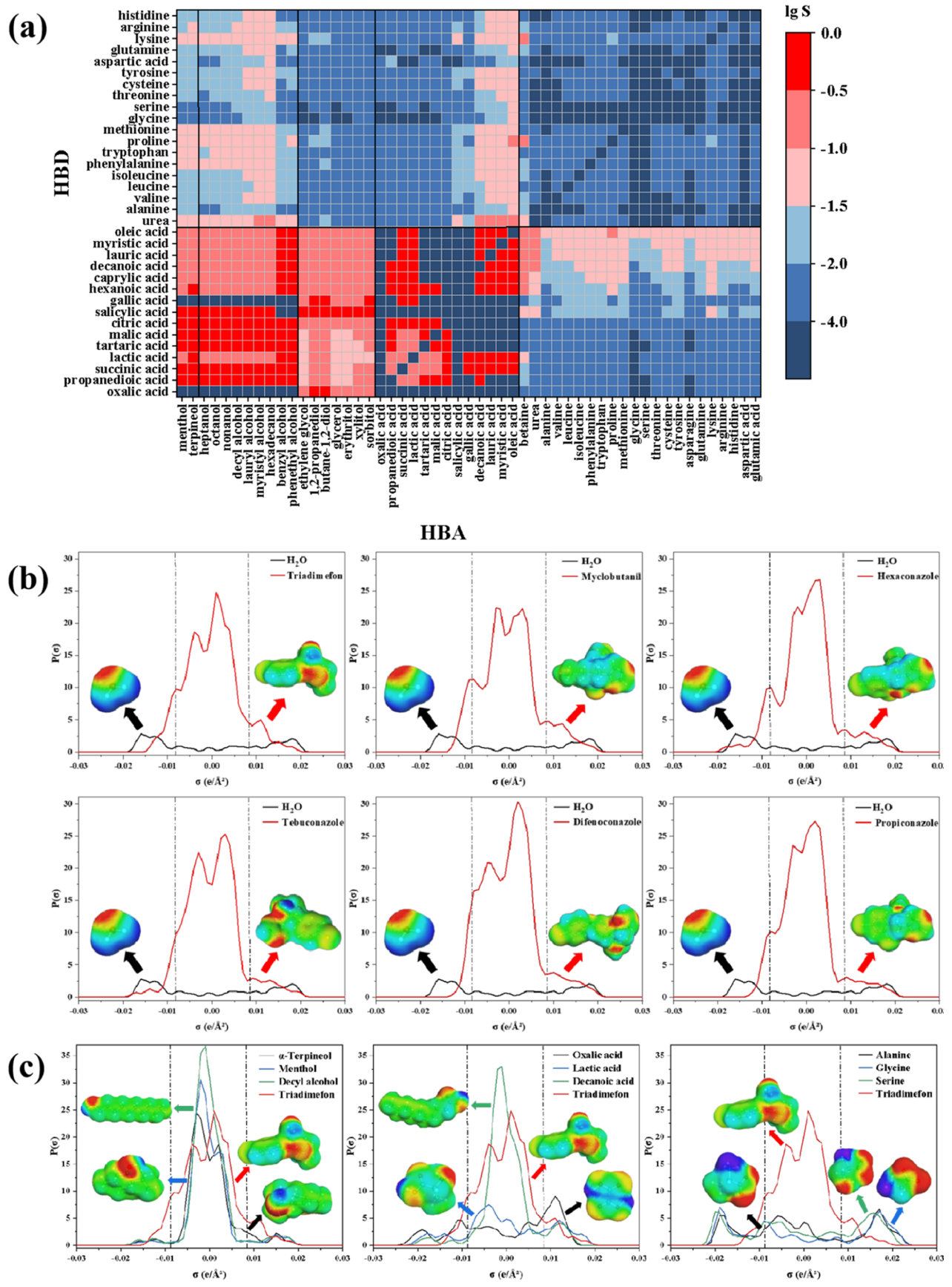
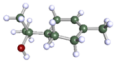
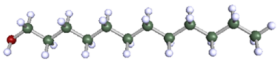
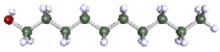
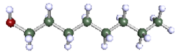
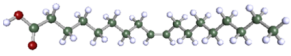
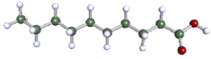
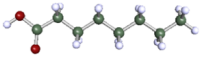
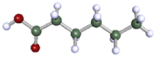
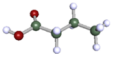


Figure 1. COSMO-RS solubility calculation results and distribution of σ-profiles for compounds: (a) Heat map illustrating the predicted solubility of TDF in 1768 NADESs; (b) σ-distribution plots of 6 TFs compared to water; (c) σ-distribution plots of TDF with distinct NADESs fractions.

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Table 1. Components of Successfully Prepared NADESs and Related Information

	Composition	Abbreviation	CAS	Chemical configuration
HBA	Menthol	Men	89-78-1	
	Terpineol	Ter	98-55-5	
	Lauryl alcohol	LAch	112-53-8	
	Decyl alcohol	DAch	112-30-1	
	Octanol	Oct	111-87-5	
HBD	Lactic acid	LA	50-21-5	
	Oleic acid	OlA	112-80-1	
	Decanoic acid	DA	334-48-5	
	Octanoic acid	OcA	124-07-2	
	Hexanoic acid	HeA	142-62-1	
	Butyric acid	BuA	107-92-6	

containing moistened filter paper and germinated under dark conditions. Ten germinated wheat seeds were selected and transplanted into plastic pots filled with 50 g of soil, supplemented with an appropriate amount of water. The pots were incubated and cultured in a controlled environment at a temperature of 25 ± 1 °C and a humidity of 65 ± 1% for 14 h per day and 10 h per night alternately. When the wheat plants reached the three-leaf and one-heart stage, they were sprayed with a 200 mg/L solution of TDF-NADESs and EC. Deionized water was used as a blank control. After 3 days of treatment, various plant traits, including stem length (cm), root length (cm), dry and fresh weights of stems (mg), dry and fresh weights of roots (mg), and chlorophyll content (mg/g), were measured to assess the safety and impact on wheat growth.

Evaluation of Cytotoxicity. Cytotoxicity was assessed using the CCK-8 assay. L929 cells obtained from the Shanghai Cell Bank of the Chinese Academy of Sciences were inoculated into 96-well plates and cultured for 24 h. After the culture medium was discarded, the cells were rinsed twice with phosphate buffered saline (PBS). Equal volumes of TDF-NADESs and EC at 10 mg/kg were added to the experimental group, while the control group received only the culture medium. After 24 h of continued incubation in the dark, the culture solution was removed, and the wells were rinsed twice with PBS. 100 μL of CCK-8 reagent was added to each well and placed in the incubator for 2 h. The absorbance at 450 nm was measured by using an enzyme labeling instrument. All cytotoxicity assays were performed in quintuplicate, and the cell survival rate was calculated by eq 8.

$$\text{relative cell viability \%} = \frac{OD_{\text{experimental group}} - OD_{\text{background}}}{OD_{\text{control group}} - OD_{\text{background}}} \times 100\% \quad (8)$$

where $OD_{\text{background}}$ is the absorbance of the multiwell plate medium itself.

RESULTS AND DISCUSSION

COSMO-RS Screening Results. A comprehensive screening process employing COSMO-RS was conducted using a diverse set of 53 organic alcohols, terpenoids, organic bases, organic amines, and amino acids as HBAs and 34 organic amines and amino acids as HBDs (Table S2). These components were combined in a 1:1 molar ratio, resulting in the generation of 1768 NADESs. The solubilization potential of these NADESs toward TDF was predicted, utilizing TDF as a model drug (Figure 1a). The heat map visually represents the solubility capacity, with color transitions from red to blue indicating a gradual decrease in solubilization efficacy. Notably, NADESs with terpenoids and long-chain alkanols as HBAs, coupled with polyacids and long-chain alkanoic acids as HBDs, demonstrated superior solubilization capabilities. Conversely, NADESs containing amino acids exhibited diminished solubilization efficiency.

Based on the shielding charge density (σ), the σ-profile can be categorized into three regions, where $-0.0082 < \sigma < 0.0082$ e/Å^2 represents the nonpolar region, $\sigma > 0.0082$ e/Å^2 corresponds to the HBA region, and $\sigma < 0.0082$ e/Å^2 designates the HBD region.[32,52] Generally, compounds exhibiting pronounced peaks in the nonpolar region tend to exhibit enhanced solubility when paired with NADESs sharing a similar peak distribution. Furthermore, if the target compound contains an HBA group, it tends to exhibit a higher affinity for NADESs with an HBD group and vice versa.[1,53] Analyzing the distribution curve of σ-profiles (Figure 1b), it is evident that TFs are nearly insoluble in water due to the predominance of their σ-values in the nonpolar region with

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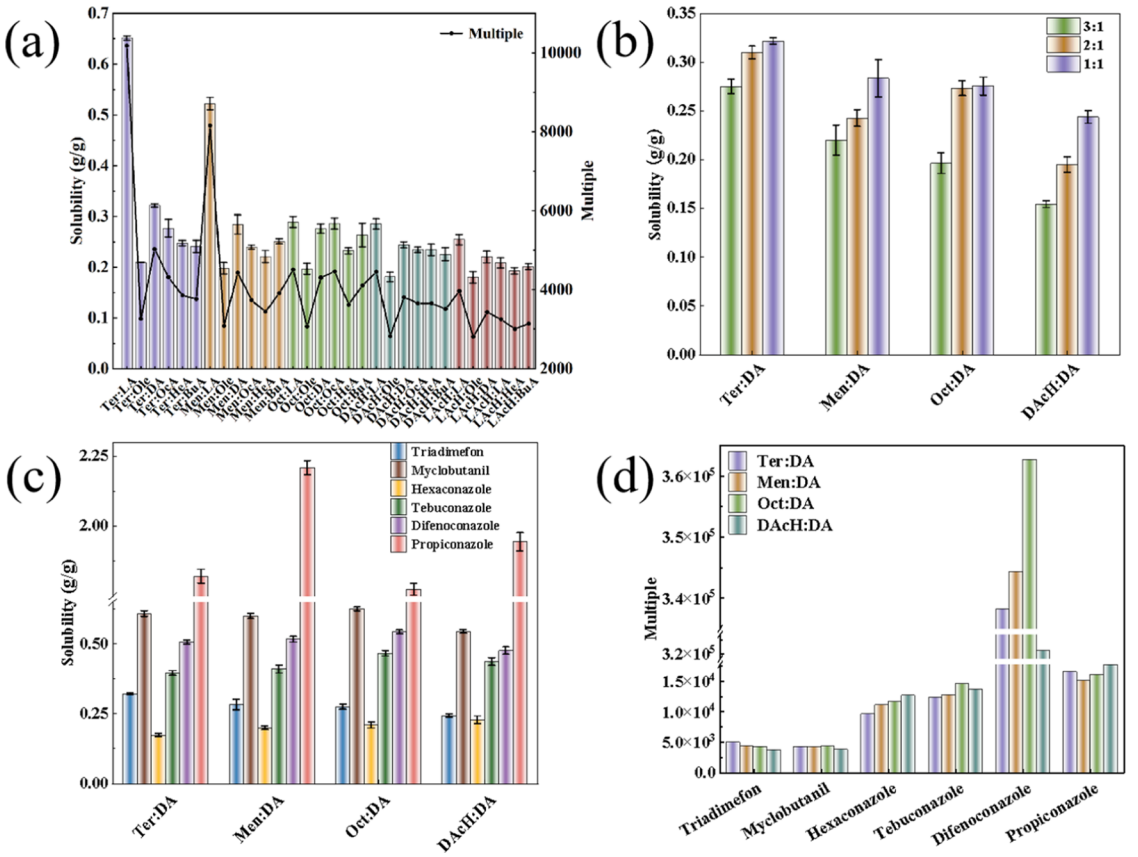


Figure 2. Solubility of TFs in NADESs. (a) The solubility of TDF in NADESs and solubilization factor in water. (b) The solubility of TDF in DA-based NADESs with varied ratios. (c) The solubility of DA-based NADESs for six TFs. (d) Multiples of solubility for six TFs relative to water.

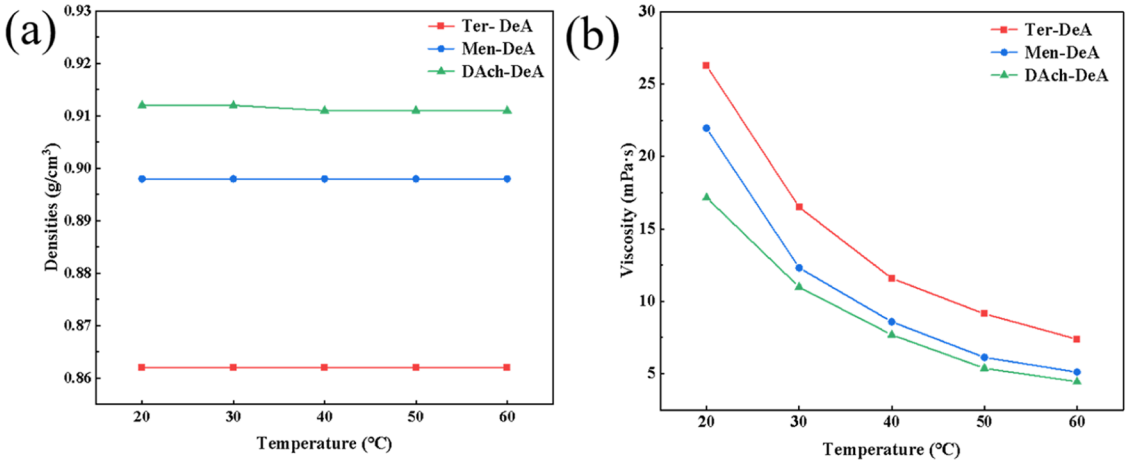


Figure 3. Trends of density (a) and viscosity (b) of NADESs at different temperatures.

Table 2. Physical Properties of NADESs

NADESs	π^*	α	β	T_m (°C)	n_D	R_D (cm^3/mol)
Ter-DeA	0.360	0.521	0.673	−0.7	1.456	210.72
Men-DeA	0.395	0.564	0.605	22.2	1.447	222.79
DAch-DeA	0.384	0.549	0.561	15.1	1.436	214.04

high intensity. In contrast, water, being a polar molecule, primarily exhibits peaks in the low-intensity HBD and HBA regions.

The calculated fractions of NADESs with superior solubilization capabilities for TDF, such as those containing terpenoids, long-chain alkanols, polyacids, and long-chain alkanoic acids, display prominently peaked distributions in the nonpolar region (Figure 1c), aligning closely with the distribution of the target drug. Conversely, amino acid-based NADESs, characterized by σ values mainly distributed in the low-intensity HBD and HBA regions, demonstrate a weaker solubilization capacity for TDF. Additionally, the σ distribution of TDF reveals small packet peaks in the HBA region, indicating a prevalence of HBA groups compared to the number of donor groups. Consequently, the strategic selection

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Table 3. Basic Properties of NADESs at the Ratio of Ter and DA from MD Simulation

	diffusion coefficient (10^{-5} cm/s)				free energy of solvation (kcal/mol)		
Ter/DA	DA	(*R*)-Ter	(*S*)-Ter	density (kg/cm^3)	$\Delta G_{Coulomb}$	ΔG_{vdW}	ΔG_{total}
1:1	0.0306 ± 0.0035	0.0319 ± 0.0004	0.0300 ± 0.0012	0.919 (0.898[a])	−1.72	−13.09	−14.81
2:1	0.0217 ± 0.0016	0.0205 ± 0.0017	0.0201 ± 0.0025	0.927 (0.931[a])	−1.56	−13.16	−14.72
3:1	0.0160 ± 0.0020	0.0147 ± 0.0021	0.0143 ± 0.0037	0.931 (0.937[a])	−1.57	−13.03	−14.60

[a]Experimental result.

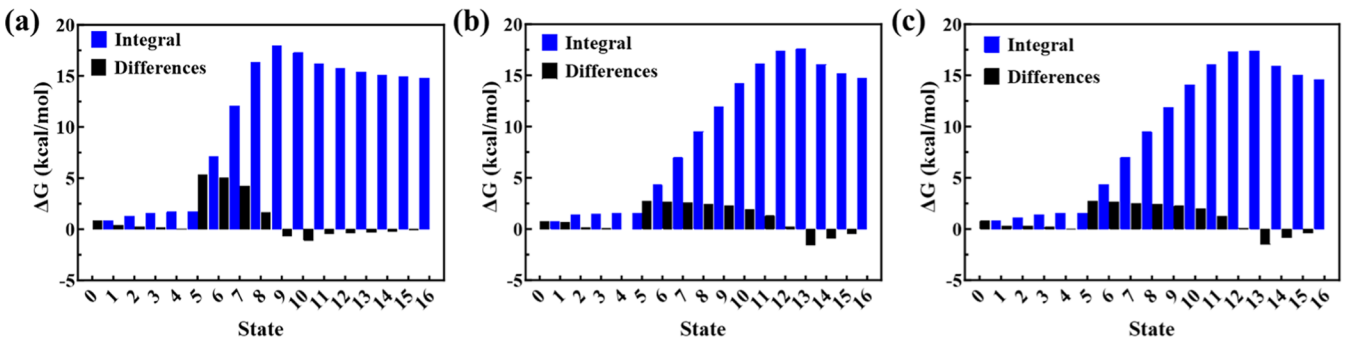


Figure 4. Free energy changes of TDF during the FEP decoupling process in Ter/DA at ratio 1:1 (a), 2:1 (b), and 3:1 (c).

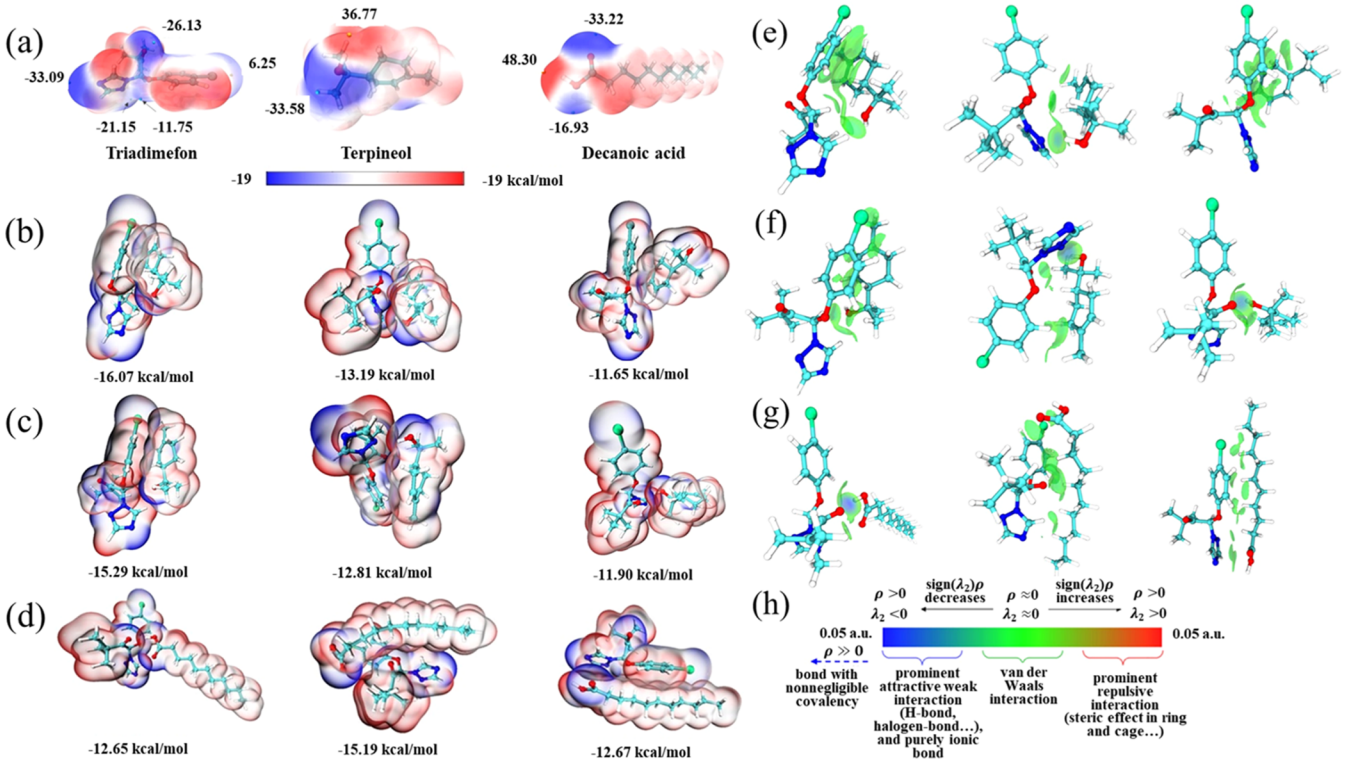


Figure 5. Quantum chemical calculations of the interaction mode and strength of TDF with solvent molecules. (a) ESP-mapped vdW surface of each molecular. Superposition of ESP-mapped vdW surfaces of the TDF with *R*-Ter (b), *S*-Ter (c), and DA (d) monomers in the dimer structure. Independent gradient model based on Hirshfeld partition (IGMH) maps of binding geometries of TDF and (e) *R*-Ter, (f) *S*-Ter, (g) DA, and (h) common interpretation of coloring method of mapped function.

of solvent components, such as organic acids with HBD groups, can effectively enhance its dissolution.

Experimental Solubility. Thirty NADES combinations with quality solubility capacity were screened by COSMO-RS. These NADESs were prepared by the heating method and kept in a clarified liquid state at 25 °C. The specific compositions are shown in Table 1. The actual solubility of these NADESs for TFs was determined by HPLC (Table S1). Figure 2 illustrates that the NADESs, identified through COSMO-RS screening, indeed exhibited an excellent dissolution capability for TDF as a model drug. This validation supports the predictive feasibility of COSMO-RS. Consequently, COSMO-RS could serve as a valuable tool for solvent design and screening, facilitating the selection of desirable NADESs from a vast array of combinations.[29] This feature is particularly crucial

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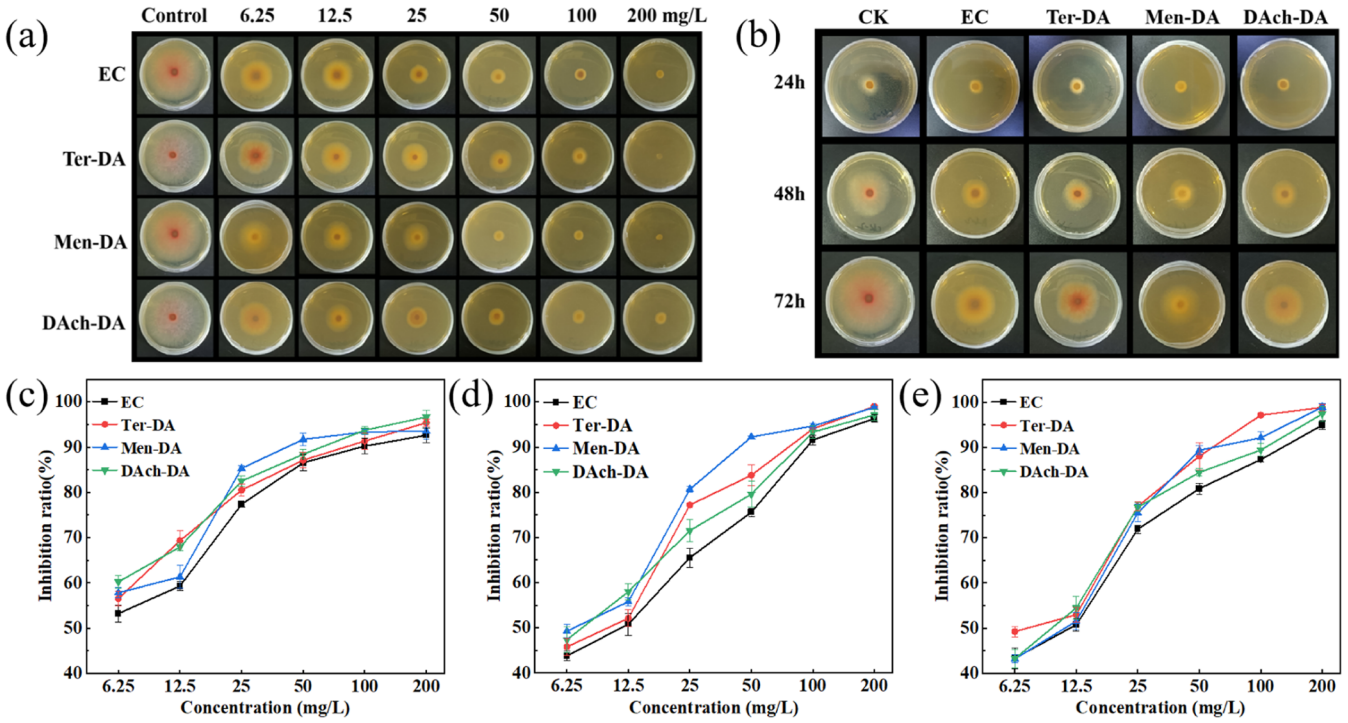


Figure 6. In vitro activity of TDF-NADESs against *F. graminearum.* Inhibition colony images at different concentrations (a) and times (b); inhibition rates at 24 h (c), 48 h (d), and 72 h (e).

Table 4. Toxicity Regression Equations and EC_{50} Values against *F. graminearum*

time (h)	formulations	toxicity regression equation	R^2	EC_{50} (mg/L)
24	EC	$y = 0.985x + 4.290$	0.965	5.252
	Ter-DA	$y = 0.991x + 4.416$	0.995	3.864
	Men-DA	$y = 1.037x + 4.314$	0.975	4.589
	DAch-DA	$y = 1.077x + 4.370$	0.994	3.848
48	EC	$y = 1.337x + 3.617$	0.968	10.826
	Ter-DA	$y = 1.605x + 3.442$	0.968	9.353
	Men-DA	$y = 1.542x + 3.679$	0.971	7.197
	DAch-DA	$y = 1.324x + 3.770$	0.977	8.482
72	EC	$y = 1.203x + 3.817$	0.986	9.617
	Ter-DA	$y = 1.658x + 3.461$	0.975	9.443
	Men-DA	$y = 1.630x + 3.401$	0.968	9.584
	DAch-DA	$y = 1.352x + 3.720$	0.977	8.842

in the pursuit of green and efficient experimental methodologies.

The actual solubility measurements are presented in Figure 2a, revealing a notable enhancement in NADES solubility, ranging from approximately 2.80×10^3 to 1.02×10^4-fold compared to TDF solubility in water. Among these, Ter/LaA 1:1 and Men/LaA 1:1 exhibited the highest solubilities, reaching 0.65 and 0.52 g/g, respectively, representing enhancements of 1.02×10^4 and 8.16×10^3-fold, respectively. Furthermore, DA-based NADESs also demonstrated outstanding solubility with values of 0.24−0.32 g/g. Relative to water, these solubilization multiples ranged from 3.43×10^3 to 5.02×10^3 times.

Considering the pronounced acidic nature of LaA (Table S3), potentially limiting its practical applicability, our focus shifted to investigating NADESs based on DA in subsequent analyses. The solubilization capability of various DA-based NADESs, exhibiting different molar ratio variations, was initially explored for TDF. As depicted in Figure 2b, the solubilization efficacy of TDF increased gradually with increasing content of DA in the NADESs. This phenomenon could be attributed to the ability of DA to enhance the lipophilicity of the NADESs, thereby improving solubilization. The specific dissolution mechanism will be detailed in Exploration of the Dissolution Mechanism. Furthermore, the robust solubility observed in COSMO-RS screened NADESs for TDF extended to other TFs, namely, myclobutanil, hexaconazole, tebuconazole, difenoconazole, and propiconazole. Solubility enhancements for these five TFs ranged from 3.8×10^3 to 3.6×10^3 times relative to their solubility in water (Figure 2c,d). The NADESs specifically designed and screened for TDF exhibited remarkable solubilization abilities for other TFs as well. This underscores the efficiency of COSMO-RS in specific design and screening, streamlining the research process, and conserving valuable time and resources.

Characterization of NADESs. Density and viscosity are pivotal physical attributes of NADESs. In this investigation, the actual viscosities of Ter-DA, Men-DA, and DAch-DA were initially determined. As depicted in Figure 3a, the viscosities of the three NADESs at 20 °C are exceptionally low, registering viscosity values in the range of 17.16−26.28 mPa·s. These values significantly fall below the viscosity of common DESs (viscosity values >100 mPa·s).[54] The ultralow viscosity of studied NADESs greatly enhances their applicability and facilitates a sufficient dissolution process. Many NADESs often necessitate the addition of water or other solvents due to their high viscosity, impacting mass transfer rates and limiting their practical utility. However, higher viscosity is frequently associated with a more extensive hydrogen bonding network. In conjunction with the solubility results, it becomes evident that the order of solubility for TDF at room temperature is Ter-DA > Men-DA > DAch-DA, while the order of viscosity is

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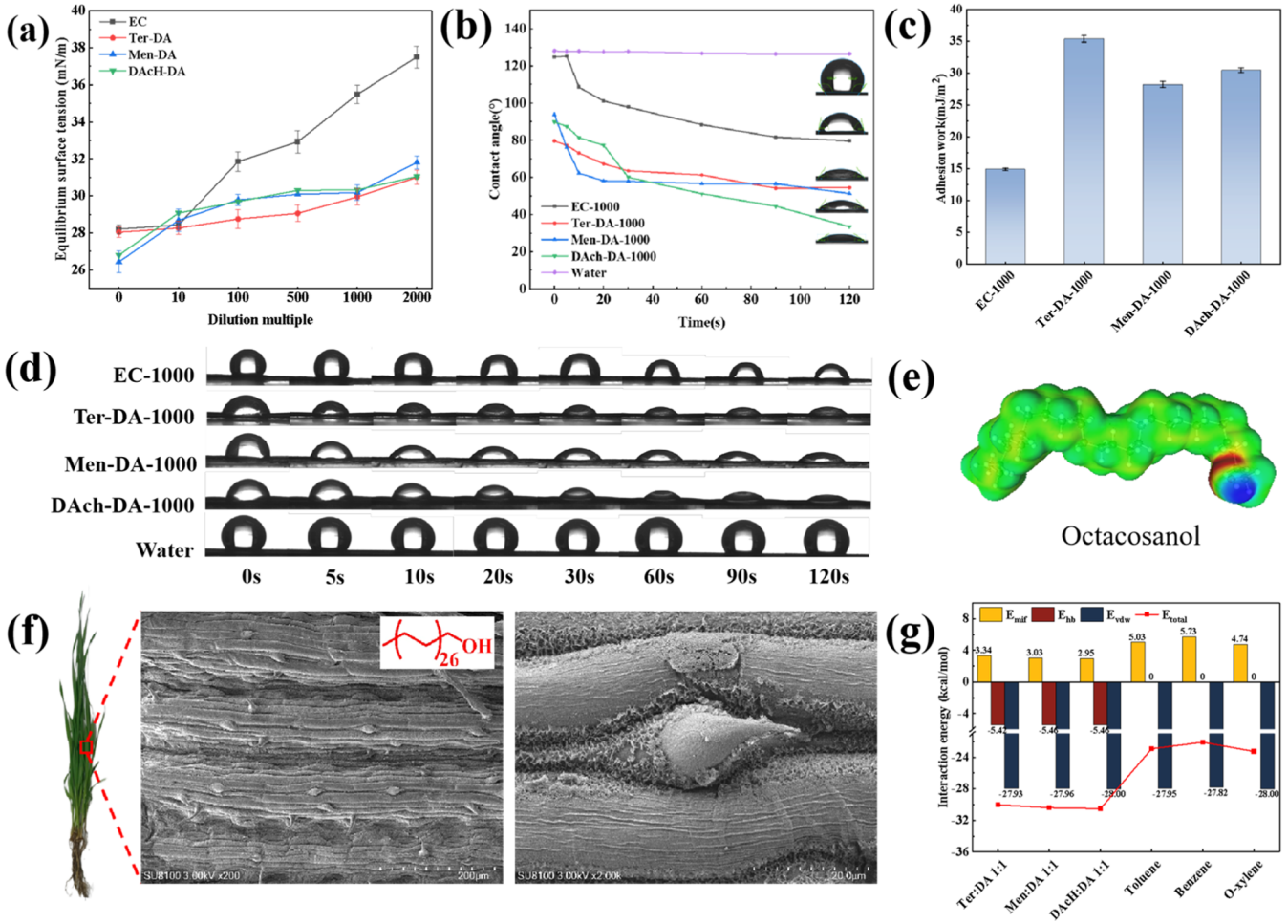


Figure 7. Adhesion of TDF-NADESs on wheat leaves. EST (a), DCA on wheat leaf blades (b), work of adhesion (c), contact angle droplet photographs of the fluids (d), the COSMO modeling of octacosanol (e), SEM of wheat leaf blades surfaces (f), and NADESs and conventional solvent interaction with wheat leaf surface (g).

Ter-DA < Men-DA < DAch-DA, indicating that, under the given conditions, lower viscosity correlates with superior dissolution capacity. Furthermore, the viscosities of NADESs exhibit a gradual decrease with rising temperature. Elevated temperatures augment intermolecular motion, diminishing intermolecular interaction forces and consequently reducing viscosity.

Additionally, as illustrated in Figure 3b, the densities of three NADESs were all lower than that of water at 20 °C, consistent with the general observation that the density of most hydrophobic DESs is lower than that of water.[54] The lower density may be attributed to the presence of DA, increasing compressibility and resulting in augmented free volume and a decrease in density. In addition, the density trend of NADESs at room temperature was opposite to the solubility capacity, suggesting that a lower density is more favorable to promote the solubilization of TDF. Similarly, with the rise in temperature, weakened intermolecular interaction forces and expanded intermolecular voids in NADESs lead to an increase in molecular volume and a subsequent decrease in density.[55]

The melting points (T_m) of NADESs were consistently lower than those of their individual components. The substantial interactions between HBAs and HBDs contributed to the reduced T_m observed in eutectics.[56] A lower T_m increased the practical application range of NADESs. Table 2 illustrates that NADESs T_m was influenced by various factors, including different HBDs, the mode of interaction with decanoic acid, and lattice energy. Furthermore, the refractive index, a crucial physical property of NADESs, provided insights into molecular interactions between HBAs and HBDs in the NADES composition.[56] The molar refractive indices of the three NADESs in Table 2 align with their solubility capabilities. As refractive indices decreased, the strengthening of hydrogen bonding interactions between components impacted the solubility of TFs.

Polarity stands out as a critical property of NADESs, playing a pivotal role in solvent solubility.[57] One widely adopted measure of polarity is the Kamlet−Taft solute discoloration parameter, which gauges the polarity/polarizability, HBD capacity, and HBA capacity of NADESs through the measurement of π^*, α, and β values, respectively, using a dye indicator.[42] As outlined in Table 2, correlating polarity parameters with the observed solubilization ability for TDF, a relationship between the solubilization capacity for TFs and the Kamlet−Taft parameter can be established; specifically, higher HBA ability (β value) with lower polar π^* value and α value indicates a more potent solubilization capacity.

Exploration of the Dissolution Mechanism. To delve into a more comprehensive understanding of the solubilization of TDF in NADESs, MD simulations were conducted.[58] The

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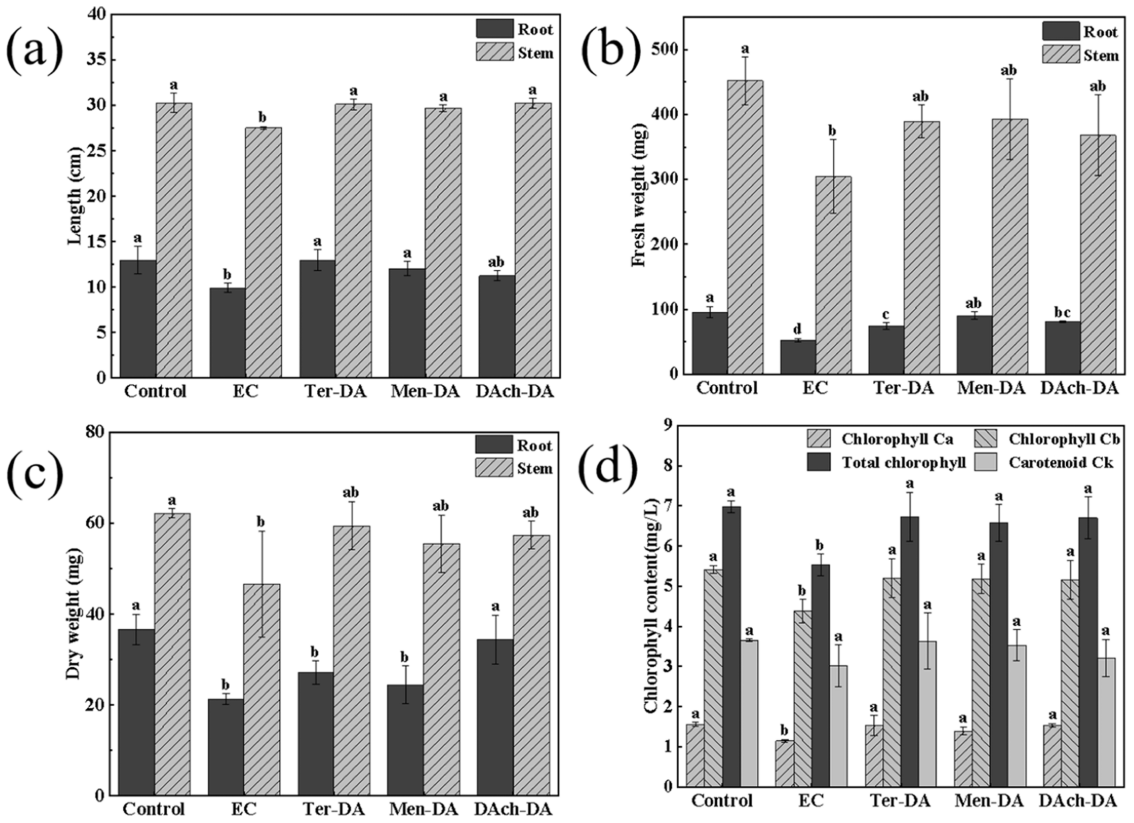


Figure 8. Biosafety of TDF-NADESs under field application concentration (200 mg/L) treatments. Roots and stems length (a), roots and stems fresh weight (b), roots and stems dry weight (c), and leaf chlorophyll content (d).

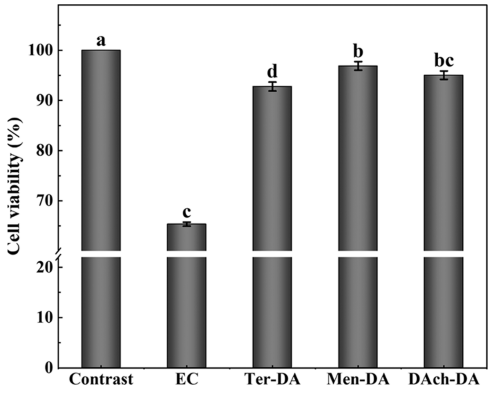


Figure 9. Cytotoxicity evaluation of TDF-NADESs.

primary focus was on exploring the fundamental properties of NADESs composed of DA and Ter. Generally, the mobility of NADESs undergoes gradual changes, corresponding to variations in the ratio. In essence, higher DA ratios result in lower intermolecular interactions with Ter, leading to increased mobility. The optimization of this ratio holds potential for practical solvent carrier applications. Simulations were performed for Ter-DA with molar ratios of 1:1, 2:1, and 3:1. Diffusion coefficients for each substance in these three eutectic solvents were calculated, revealing that the mixtures with a 1:1 ratio exhibited the highest mobility (Table 3). Additionally, the densities of these solvents showed relatively small deviations from the experimental results, validating the force field parameters for these molecules.

Subsequently, the determination of free energies of solubilization of TDF in each solvent was carried out through free energy perturbation (FEP) calculations to comprehend the impact of differences on solubility (Figure S1). In alignment with the experimental findings, the free energy of TDF dissolution gradually decreased with an increase in the Ter ratio in the solvent. Specifically, the Ter and DA ratio of 1:1 yielded the lowest free energy of dissolution (Table 3), signifying optimal solubilization of TDF in this solvent. In FEP calculations, Coulombic and van der Waals interactions were decoupled sequentially (Figure 4), elucidating the contribution of each interaction. The analysis revealed that van der Waals interactions predominantly governed dissolution; however, the decline in free energy of dissolution from 1:1 to 2:1 was primarily steered by Coulombic interactions.

In-depth exploration of the interaction mode and strength between TDF and solvent molecules was conducted through quantum chemical calculations on a series of solvent−solvent dimers. These dimers were constructed via conformational searching for each monomer. Quantitative analysis of the molecular surface electrostatic potential (ESP) revealed consistently low values, even at the maximum or minimum points of the ESP (Figure 5a). Binding energies were computed, and three geometries representing the most stable binding configurations between TDF and each solvent molecule were selected for further analysis. The superposition of van der Waals surfaces of TDF and solvent molecule monomers in the dimeric structure is depicted in Figure 5b−d. Notably, the ESP-negative region around the TDF ideally overlapped with the ESP-positive region around the solvent molecule, whether it was DA or Ter.

Despite TDF possessing typical HBAs such as the 4-position of the triazole ring and the carbonyl group, favorable binding

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geometries based on hydrogen bonding were not achieved. Conversely, dispersive interactions between the aromatic ring and the alkyl group in TDF provided an optimal binding posture for various solvent molecules. These findings provide further support for the dominance of van der Waals interactions in the solvation process. To elucidate the weak interactions between solute and solvent, independent gradient modeling between fragments based on Hirshfeld division analysis was performed for each selected complex (Figure 5e−g).[59] Analyzing the distribution of the major van der Waals interaction region, represented by the green isosurface, unequivocally demonstrated the prevalence of van der Waals interactions in the solvation process. Furthermore, compared to TDF and Ter, the darker blue region around the center of the isosurface corresponding to the O−H···O interaction between TDF and DA indicated a stronger hydrogen bond. This result suggests a differential contribution of Coulombic interactions to solvation-free energy, particularly, in the transition from a 1:1 to a 2:1 ratio.

Stability Assessment. Pesticides necessitate a certain duration between their production and utilization. To ensure product quality, the stability of AIs in formulations is typically evaluated through hot storage stability tests. The thermal storage stability of TDF, along with myclobutanil, hexaconazole, tebuconazole, difenoconazole, and propiconazole, was assessed when it was dissolved in Ter-DA, Men-DA, DAch-DA, and Oct-DA using the CIPAC method. Both high and low concentrations maintained the stability of AIs, with content degradation before and after storage remaining below 5%. The formulations based on NADESs demonstrated the capacity to sustain the stability of product activity over a defined period (Figure S2).

In Vitro Activity Evaluation. *F. graminearum* stands as the primary pathogen responsible for wheat scab, a highly destructive and challenging-to-control disease in agricultural production. Varied concentrations of TDF in Ter-DA, Men-DA, DAch-DA, and commercial EC were subjected to inhibitory activity assessments against *F. graminearum*, with results illustrated in Figure 6. The colony images in Figure 6a, captured at 72 h, demonstrate effective inhibition of *F. graminearum* growth across concentrations ranging from 6.25 to 200 mg/L. Figure 6b showed images of colony growth at different times under treatment with the concentration applied in the field (200 mg/L). The concentration-dependent inhibitory effects of TDF formulations were evident, showing a substantial increase in inhibition rates with increasing concentrations (Figure 6c−e). TDF-NADESs exhibited significantly higher inhibitory effects compared to EC. For instance, at 24 h, the inhibition rate for TDF EC was 53.2% at the lowest concentration (6.25 mg/L) and 92.6% at the highest concentration (200 mg/L). In contrast, TDF-NADESs achieved rates of 60.2 and 93.7%, respectively, indicating increased bactericidal activity in the presence of NADESs.

Furthermore, the inhibitory effect at concentrations below 100 mg/L gradually decreased with an extended treatment time. However, at higher concentrations, not only did the inhibitory effect remain unaffected but it also exhibited an enhancement. EC_{50} values at 24 h for EC, Ter-DA, Men-DA, and DAch-DA were 5.252, 3.864, 4.589, and 3.848 mg/L, respectively, indicating superior activities of TDF-NADES formulations (Table 4). In conclusion, TDF-NADES formulations exhibited excellent in vitro bacterial inhibitory activities, showcasing significant potential for practical applications in agriculture.

Adhesion of TDF-NADESs on Wheat Leaves. The adherence of droplets to wheat leaves is pivotal for effective pesticide distribution and utilization.[60] Figure 7a depicts the EST of TDF-NADESs and EC drops at varying dilutions, with field applications typically conducted at a 1000-fold dilution. At this concentration, the EST of the TDF-NADESs liquid was 14.5−15.6% lower than EC at the field application dilution, proving that the presence of NADESs contributed to a reduction in the EST of the formulation system. Additionally, the wettability of the formulation system on wheat leaves was evaluated through DCA measurements. As shown in Figure 7b, wheat leaf surfaces exhibited high hydrophobicity, evidenced by water's minimal wetting, with a DCA changing from 128 to 127° over 120 s. At a 1000-fold dilution, NADESs-based formulations exhibited 31.6−58.1% lower DCA, facilitating enhanced wetting and faster spreading on wheat leaves compared to EC.

As shown in Figure 7c, the adhesion work of EC was 14.9 mJ/m^2, while Ter-DA, Men-DA, and DAch-DA exhibited higher values of 35.3, 28.2, and 30.4 mJ/m^2, respectively. This underscores that TDF-NADESs formulations exhibited superior adhesion to hydrophobic wheat leaves, enhancing solution utilization.

The hydrophobic component of wheat leaves primarily comprises a waxy layer, predominantly composed of sterols with chain lengths ranging from C20 to C32, where C28 is the dominant component. SEM revealed a microstructure with numerous folds, trichome structures, and a dense arrangement of waxy lamellar crystals, accentuating the leaf hydrophobicity (Figure 7f). The intermolecular interactions between C28 sterols and NADESs and conventional solvents (benzene, toluene, and xylene) were calculated using a COMO-RS simulation (Figure 7e,g). The results indicated that NADESs exhibited stronger van der Waals forces and readily formed intermolecular hydrogen bonds with C28 sterols. This suggests that TDF-NADESs formulations hold promising applications in wheat and similar crops. Leveraging the morphology, microstructure, and composition of target crop leaves, cosolvents for active ingredients can be designed and screened using COSMO-RS, providing a simplified and effective approach for future pesticide formulation design and preparation.

Biosafety of Assessment of TDF-NADESs. *Impact on Wheat Growth.* Wheat, being highly susceptible to environmental variations, experiences growth inhibitory effects due to external environmental resistance and stress factors.[32] Assessing the biosafety of TDF-NADESS involves measuring wheat growth indicators. Figures 8 and S1 depict the influence of EC and NADESs formulations on wheat growth at the field application concentration. Seven days postapplication, NADESs formulations treatments displayed no significant inhibitory effects on wheat, with no considerable impact on root length, stem dry weight, and fresh weight. And Ter-DA and DAch-DA affected the fresh weight of wheat root slightly, indicating greater sensitivity of wheat roots to environmental changes than aerial parts. This sensitivity might be attributed to the predominant retention of the sprayed solution in the soil, exerting a more pronounced effect on the root growth. In contrast, EC treatment significantly inhibited wheat growth indicators, encompassing both root and stem lengths as well as fresh and dry weight. Additionally, the chlorophyll content

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analysis in wheat leaves indicated that EC inhibited chlorophyll production, while NADESs formulations exhibited negligible effects on the assessed indexes, indicating that the NADESs formulation did not inhibit wheat photosynthesis.

Therefore, compared to EC, the application of NADES formulations mitigated the impact on each growth parameter of wheat, ensuring minimal disruption to photosynthesis and highlighting a high level of safety for wheat.

Cytotoxicity Assessment. Inevitable exposure to pesticides during application necessitates an evaluation of cytotoxicity. The CCK-8 method was employed to test mouse fibroblasts L929, with the aim of assessing the biosafety of NADESs formulations. Test results (Figure 9) revealed that, in comparison to the control group, the cell survival rates for NADES formulations at a concentration of 100 mg/L were notably high 24 h post-treatment of L929, registering at 92.8−96.9%. Notably, these rates surpassed that of the EC, which stood at 65.4%. This substantiates the biocompatibility of NADESs formulations, affirming their practical and safe handling by humans in field applications.

CONCLUSIONS

In this comprehensive investigation, COSMO-RS was utilized for the innovative design and high-throughput screening of novel NADESs with a superior solubilizing capacity for TDF. The reliability of COSMO-RS computational screening was corroborated by determining the actual solubility. The solubility of screened NADESs for TFs was enhanced by 2.8 × 10^3 to 3.6 × 10^5 times relative to water. This substantiated the universality of the screened NADESs against TDF for various TFs. Combining MD simulations, quantitative calculations, and properties, the critical factors and sites affecting the solvency of NADESs on TFs were identified as the hydrogen bonding between the triazole ring and hydroxyl group as well as the van der Waals force between benzene rings, which provided a foundational theoretical framework for future solvent design and screening endeavors. In addition, the properties of NADESs, such as viscosity, density, and polarity, also have a certain influence on their solubilization ability. Moreover, compared to conventional EC, the creation of eco-friendly TDF-NADESs formulation demonstrated enhanced in vitro inhibitory activity, superior adhesion, wetting abilities of wheat leaves, and lower EC_{50} against *F. graminearum*. The interaction between NADESs components and leaf surface substances was systematically explored through calculations and characterization. Additionally, the pesticide systems of NADESs formulations exhibited diminished effects on wheat crop growth, lower cytotoxicity, and commendable biosafety. Leveraging the advantageous features of NADESs, such as natural components, high solubility, design flexibility, and the capacity to enhance pesticide utilization, holds tremendous promise for elevating pesticide efficiency, mitigating environmental pollution, and ushering in a new era of sustainable and efficient pesticides.

ASSOCIATED CONTENT

Data Availability Statement

Data will be made available on request.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acssuschemeng.4c03110.

Specific operation and parametrization of solvent simulation; table of TFs information and HPLC detection methods (Table S1); table of information on the NADESs components (Table S2); table of dissociation constant (pK_a) value of the hydrogen bond donor (Table S3); schematic diagram of the solvation-free energy calculation (Figure S1); diagram of thermal storage stability of six TFs in NADESs (Figure S2); photograph of wheat growth status (Figure S3) (PDF)

AUTHOR INFORMATION

Corresponding Authors

Fengmao Liu − *Innovation Center of Pesticide Research, Department of Applied Chemistry, College of Science, China Agricultural University, Beijing 100193, China;* orcid.org/0000-0002-1958-7230; Phone: 0086-10-62731978; Email: lfm2000@cau.edu.cn; Fax: +0086-10-62733620

Qingrong Peng − *Innovation Center of Pesticide Research, Department of Applied Chemistry, College of Science, China Agricultural University, Beijing 100193, China;* Phone: 0086-10-62731978; Email: qrpeng@cau.edu.cn; Fax: +0086-10-62733620

Authors

Juan Wang − *Innovation Center of Pesticide Research, Department of Applied Chemistry, College of Science, China Agricultural University, Beijing 100193, China*

Xianzhao Zhang − *Innovation Center of Pesticide Research, Department of Applied Chemistry, College of Science, China Agricultural University, Beijing 100193, China*

Zhiqiang Gao − *Innovation Center of Pesticide Research, Department of Applied Chemistry, College of Science, China Agricultural University, Beijing 100193, China*

Peirong Zhang − *Innovation Center of Pesticide Research, Department of Applied Chemistry, College of Science, China Agricultural University, Beijing 100193, China*

Wenzhuo Wang − *Institute for the Control of the Agrichemicals, Ministry of Agriculture and Rural Affairs, Beijing 100125, China*

Jing Sun − *Innovation Center of Pesticide Research, Department of Applied Chemistry, College of Science, China Agricultural University, Beijing 100193, China*

Complete contact information is available at: https://pubs.acs.org/10.1021/acssuschemeng.4c03110

Notes

The authors declare no competing financial interest.

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